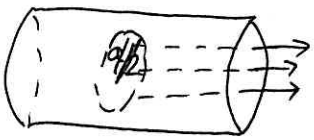
7.12



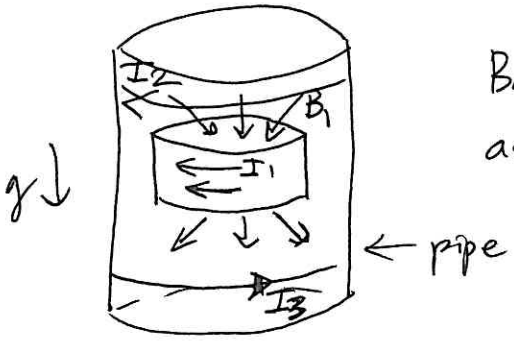
$$B(t) = B_0 \cos(\omega t)\,\hat{z}$$

$$\Phi = \int B(t)\,da = B_0 \cos(\omega t)\,\pi\left(\frac{a}{2}\right)^2$$

$$\mathcal{E}(t) = -\frac{d\Phi}{dt} = \frac{\pi a^2}{4} B_0 \omega \sin(\omega t)$$

$$I(t) = \frac{\mathcal{E}(t)}{R} = \frac{\pi a^2}{4}\frac{B_0 \omega}{R}\sin(\omega t)$$

7.14



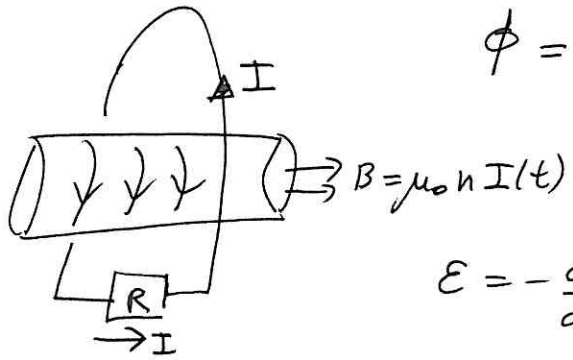
Bar magnet can be thought of as a solenoid with current I_1 flowing on its surface and producing magnet's magnetic field $\vec{B}_1$.

As it falls through the pipe, it sets of circulating eddy currents which try to minimize the change in magnetic flux: current I_2 above the bar magnet rotating the same way as I_1, and current I_3 in front of the magnet rotating opposite to I_1.

Hence both currents I_2 and I_3 produce upward force on I_1, slowing its fall down.

The effect is very similar to Example 7.6.

$\boxed{7.17}$ (a)



$$\phi = \pi a^2 \cdot \mu_0 n I(t)$$

$$\mathcal{E} = -\frac{d\phi}{dt} = -\pi a^2 \mu_0 n \underbrace{\frac{dI}{dt}}_{k}$$

$$I = \frac{\pi a^2 \mu_0 n k}{R}$$

Direction: to the right so as to reduce increasing flux $\phi(t)$.

(b) Now $I = \text{const} \Rightarrow \phi_1 = \pi a^2 \mu_0 n I$

After solenoid is removed and reinserted in the opposite direction, the direction of the flux changes

$\Rightarrow \phi_2 = -\pi a^2 \mu_0 n I \Rightarrow \Delta\phi = \phi_1 - \phi_2 = 2\pi a^2 \mu_0 n I$

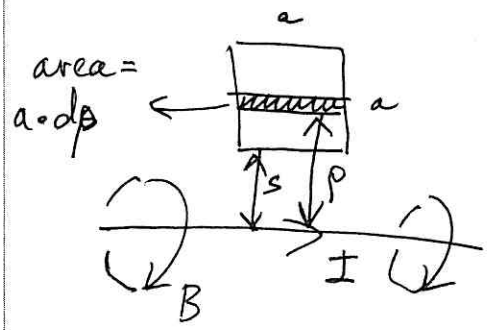
The current induced $I = \frac{\mathcal{E}}{R}$ but $I = \frac{dQ}{dt}$

$$\mathcal{E} = -\frac{d\phi}{dt}$$

$$\Rightarrow \frac{dQ}{dt} = -\frac{1}{R}\frac{d\phi}{dt} \Rightarrow |\Delta Q| = \frac{1}{R}|\Delta\phi|$$

$$|\Delta Q| = \frac{2\pi a^2 \mu_0 n I}{R}$$

$\boxed{7.18}$



Current I produces $\vec{B}(s) = \dfrac{\mu_0 I}{2\pi s}\,\hat{\varphi}$

$$\phi = \int \vec{B}\cdot d\vec{A} = \frac{\mu_0 I}{2\pi}\int_s^{s+a} \frac{a\cdot d\rho}{\rho} =$$

$$= \frac{\mu_0 I a}{2\pi}\ln\left(\frac{s+a}{s}\right).$$

<u>Check</u> Area of the loop:

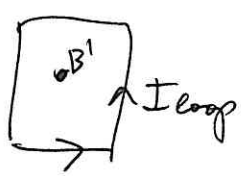
$$\int_s^{s+a} a\,d\rho = a\,\rho\Big|_s^{s+a} = a(s+a-s) = a^2 \text{ (of course)}$$

$$I_{loop} = \frac{1}{R}\,\mathcal{E} = \frac{1}{R}\left(-\frac{d\phi}{dt}\right) = -\frac{\mu_0 a}{2\pi R}\ln\left(\frac{s+a}{s}\right)\cdot\frac{dI(t)}{dt}$$

Flux ϕ is decreasing as $\dfrac{dI}{dt} = \begin{cases} -\alpha I & \text{for } t \le \frac{1}{\alpha} \\ 0 & \text{for } t > \frac{1}{\alpha}. \end{cases}$

$\Downarrow$

I_{loop} will try to support the flux $\Rightarrow$ I_{loop} flows counterclockwise in order to generate field $\vec{B}'$ pointing out of the page.



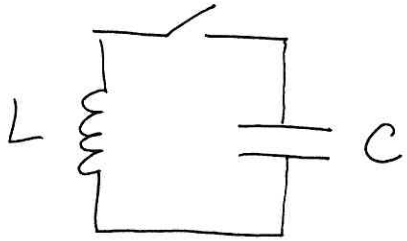
Total charge passed through the loop

$$Q = \int_0^\infty I(t)\,dt = \frac{1}{R}\int_0^\infty\left(-\frac{d\phi}{dt}\right)dt =$$

$$= -\frac{1}{R}\Big(\underbrace{\phi(t=\infty)}_{0} - \underbrace{\phi(t=0)}_{\frac{\mu_0 I a}{2\pi}\ln\left(\frac{s+a}{a}\right)}\Big) = \frac{\mu_0 I a}{2\pi R}\ln\left(\frac{s+a}{s}\right).$$

Note: solution in the <u>manual</u> is not correct! ☺

7.25



$Q = CV$ at $t=0$, and $I(t=0)=0$.

$$-L\frac{dI}{dt} = V = \frac{Q}{C}$$

$$I = \frac{dQ}{dt}$$

$$\Rightarrow L\frac{d^2Q}{dt^2} + \frac{Q}{C} = 0$$

$$\frac{d^2Q}{dt^2} = -\frac{1}{LC}Q = -\omega_0^2 Q$$

$$Q(t) = A\cos\omega_0 t + B\sin\omega_0 t$$

$$I(t) = -A\omega_0 \sin\omega t + B\omega_0 \cos\omega_0 t.$$

Initial conditions: $Q(t=0) = CV = A$

$$I(t=0) = B\omega_0 = 0$$

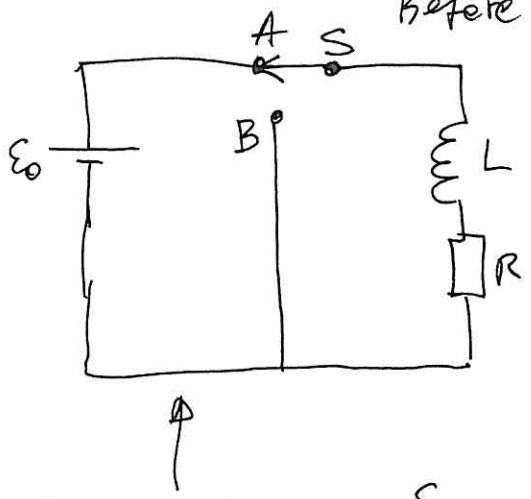
$$\Rightarrow Q(t) = CV\cos\left(\frac{t}{\sqrt{LC}}\right)$$

$$I(t) = -\frac{CV}{\sqrt{LC}}\sin\left(\frac{t}{\sqrt{LC}}\right) = -V\sqrt{\frac{C}{L}}\sin\left(\frac{t}{\sqrt{LC}}\right).$$

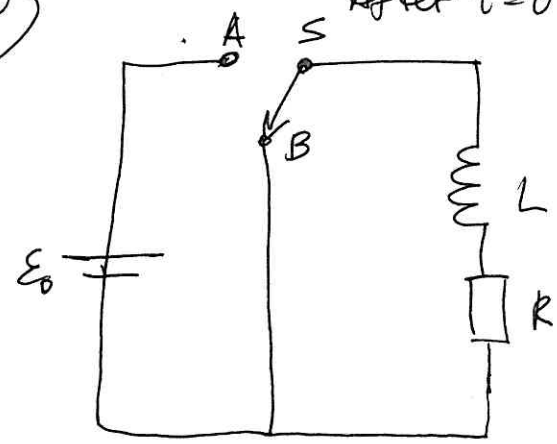
Resistance R will cause energy dissipation and decay of current with time.

7.29

Before $t=0$



After $t=0$:



steady current $I_0 = \frac{\mathcal{E}_0}{R}$

(a) At $t \geq 0$: battery is disconnected $\Rightarrow$ emf $= -L\frac{dI}{dt}$ only.

$$-L\frac{dI}{dt} = RI \Rightarrow \frac{dI}{dt} = -\frac{R}{L}I$$

$$I(t) = I_0 e^{-\frac{R}{L}t} = \frac{\mathcal{E}_0}{R} e^{-\frac{R}{L}t}$$

(b) $P = \mathcal{E}I = -L\frac{dI}{dt}I = -\frac{1}{2}L\frac{dI^2}{dt}$

Total energy $W = \int P\,dt = -\frac{1}{2}L\int_0^\infty \frac{dI^2}{dt}dt = \frac{1}{2}LI_0^2$

$$W = \frac{1}{2}L\left(\frac{\mathcal{E}_0}{R}\right)^2$$

(c) Initial energy of the inductor $= \frac{1}{2}LI_0^2$ by def.

$\Rightarrow$ coincides with W in (b), of course.